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Inventor(s)	Yoshikawa; Hiromichi

Antenna and array antenna

Abstract

An antenna includes a first resonator extending in a first plane direction; a second resonator spaced apart from the first resonator in a first direction and extending in the first plane direction; a third resonator positioned between the first resonator and the second resonator in the first direction and magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator; a reference conductor extending in the first plane direction, positioned between the first resonator and the second resonator in the first direction, and serving as a potential reference of the first resonator and the second resonator; and a feeder line connected to the first resonator. The reference conductor surrounds at least a part of the third resonator in the first plane direction.

Inventors:	Yoshikawa; Hiromichi (Yokohama, JP)
Applicant:	KYOCERA CORPORATION (Kyoto, JP)
Family ID:	1000008780913
Assignee:	KYOCERA CORPORATION (Kyoto, JP)
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Primary Examiner: Lotter; David E

Attorney, Agent or Firm: HAUPTMAN HAM, LLP

Background/Summary

RELATED APPLICATIONS

(1) The present application is a National Phase of International Application Number PCT/JP2021/045386 filed Dec. 9, 2021, which claims the benefit of priority from Japanese Patent Application No. 2021-070632, filed on Apr. 19, 2021.

TECHNICAL FIELD

(2) The present disclosure relates to an antenna and an array antenna.

BACKGROUND OF INVENTION

(3) Usually, implementation of a compact antenna with a wide band operating frequency by using a planar antenna is difficult.

CITATION LIST

Patent Literature

(4) Patent Document 1: JP 2011-155479 A

SUMMARY

Problem to be Solved

(5) The resonator element as described in Patent Document 1 has a plurality of resonance structures, and an antenna having a high degree of freedom in design is in demand.

(6) The present disclosure provides an antenna and an array antenna having a resonance structure and a high degree of freedom in design.

Solution to Problem

(7) An antenna according to the present disclosure includes a first resonator extending in a first plane direction; a second resonator spaced apart from the first resonator in a first direction and extending in the first plane direction; a third resonator that is positioned between the first resonator and the second resonator in the first direction and is magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator; a reference conductor extending in the first plane direction, positioned between the first resonator and the second resonator in the first direction, and serving as a potential reference of the first resonator and the second resonator; and a feeder line connected to the first resonator, in which the reference conductor surrounds at least a part of the third resonator in the first plane direction.

(8) An antenna according to the present disclosure includes a first resonator extending in a first plane direction; a second resonator spaced apart from the first resonator in a first direction and extending in the first plane direction; a reference conductor extending in the first plane direction, positioned between the first resonator and the second resonator in the first direction, and serving as a potential reference of the first resonator and the second resonator; a third resonator that is positioned between the first resonator and the second resonator in the first direction and is magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator; a first auxiliary reference conductor positioned between the first resonator and the reference conductor and extending in the first plane direction; a second auxiliary reference conductor positioned between the second resonator and the reference conductor and extending in the first plane direction; a first connection line path that electromagnetically connects the first resonator, the reference conductor, and the first auxiliary reference conductor; and a second connection line path that electromagnetically connects the second resonator, the reference conductor, and the second auxiliary reference conductor, in which the reference conductor surrounds at least a part of the third resonator in the first plane direction.

(9) An array antenna according to the present disclosure includes one or more antennas according to the present disclosure, in which the one or more antennas are arranged in the first plane direction.

Advantageous Effect

(10) According to the present disclosure, an antenna and an array antenna having a resonance structure and a high degree of freedom in design can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view illustrating a configuration of an antenna according to a first embodiment.

(2) FIG. 2 is a view for illustrating a radiation pattern of the antenna according to the first embodiment.

(3) FIG. 3 is a graph showing frequency characteristics of the antenna according to the first embodiment.

(4) FIG. 4 is a graph showing radiation characteristics of the antenna according to the first embodiment.

(5) FIG. 5 is a graph showing a peak gain of the antenna according to the first embodiment.

(6) FIG. 6 is a view illustrating a configuration example of an antenna according to a second

embodiment.

(7) FIG. 7 is a graph showing frequency characteristics of the antenna according to the second embodiment.

(8) FIG. 8 is a graph showing frequency characteristics of the antenna according to the second embodiment.

(9) FIG. 9 is a graph showing a peak gain of the antenna according to the second embodiment.

(10) FIG. 10 is a view for illustrating a radiation pattern of the antenna according to the second embodiment.

(11) FIG. 11 is a view for illustrating a radiation pattern of the antenna according to the second embodiment.

(12) FIG. 12 is a view for illustrating a radiation pattern of the antenna according to the second embodiment.

(13) FIG. 13 is a graph showing frequency characteristics of the antenna according to other embodiments.

DESCRIPTION OF EMBODIMENTS

(14) Embodiments of the present disclosure will be described in detail with reference to the drawings. The embodiments described below do not limit the present disclosure.

(15) In the following description, an XYZ orthogonal coordinate system is set, and the positional relationship between respective portions will be described by referring to the XYZ orthogonal coordinate system. A direction parallel to an X-axis in a horizontal plane is defined as an X-axis direction, a direction parallel to a Y-axis in the horizontal plane orthogonal to the X-axis is defined as a Y-axis direction, and a direction parallel to a Z-axis orthogonal to the horizontal plane is defined as a Z-axis direction. A plane including the X-axis and the Y-axis is appropriately referred to as an XY plane, a plane including the X-axis and the Z-axis is appropriately referred to as an XZ plane, and a plane including the Y-axis and the Z-axis is appropriately referred to as a YZ plane. The XY plane is parallel to the horizontal plane. The XY plane, the XZ plane, and the YZ plane are orthogonal to each other.

First Embodiment

(16) Configuration of Antenna

(17) A configuration of an antenna according to a first embodiment will be described with reference to FIG. 1. FIG. 1 is a view illustrating the configuration of the antenna according to the first embodiment.

(18) As illustrated in FIG. 1, an antenna 10 includes a substrate 12, a first resonator 14, a second resonator 16, a reference conductor 18, a connection line path 20, a third resonator 22, and a feeder line 30.

(19) The first resonator 14 can be arranged on the substrate 12 so as to extend in the XY plane. The first resonator 14 can be made of a conductor. The first resonator 14 can be, for example, a patch conductor formed in a rectangular shape. In the example illustrated in FIG. 1, the first resonator 14 is illustrated as a rectangular patch conductor, but the present disclosure is not limited to this. The first resonator 14 may have, for example, a linear shape, a circular shape, a loop shape, or a polygonal shape other than a rectangular shape. That is, the shape of the first resonator 14 can be freely changed depending on the design. The first resonator 14 is configured to resonate by an electromagnetic wave received from the +Z-axis direction.

(20) The first resonator 14 is configured to radiate an electromagnetic wave when resonating. The first resonator 14 is configured to radiate the electromagnetic wave toward the +Z-axis direction when resonating.

(21) The second resonator 16 can be arranged on the substrate 12 so as to extend in the XY plane at a position away from the first resonator 14 in the Z-axis direction. The second resonator 16 can be, for example, a patch conductor formed in a rectangular shape. In the example illustrated in FIG. 1, the second resonator 16 is illustrated as a rectangular patch conductor, but the present disclosure is

not limited to this. The second resonator **16** may have, for example, a linear shape, a circular shape, a loop shape, or a polygonal shape other than a rectangular shape. That is, the shape of the second resonator **16** can be freely changed depending on the design. The shape of the second resonator **16** may be the same as or different from the shape of the first resonator **14**. The area of the second resonator **16** may be the same as or different from the area of the first resonator **14**.

(22) The second resonator **16** is configured to radiate an electromagnetic wave when resonating. The second resonator **16** is, for example, configured to radiate the electromagnetic wave toward the $-Z$ -axis direction. The second resonator **16** is configured to radiate the electromagnetic wave to the $-Z$ -axis direction when resonating. The second resonator **16** is configured to resonate by receiving the electromagnetic wave from the $-Z$ -axis direction.

(23) The second resonator **16** may resonate at a phase different from that of the first resonator **14**. The second resonator **16** may be configured to resonate in a direction different from that of the first resonator **14** in the XY plane direction. For example, when the first resonator **14** is configured to resonate in the X-axis direction, the second resonator **16** may be configured to resonate in the Y-axis direction. The second resonator **16** may be configured such that the resonance direction of the second resonator **16** changes over time in the XY plane direction in response to change over time in the resonance direction of the first resonator **14**. The second resonator **16** may be configured to radiate an electromagnetic wave with a first frequency band attenuated from the electromagnetic wave received by the first resonator **14**. The reference conductor **18** reduces cancellation of a current contributing to radiation when a coupled mode relationship is established among the first resonator **14**, the second resonator **16**, and the third resonator **22**. With reference conductor **18** being present, radiation at the frequency of each coupled mode is performed.

(24) The reference conductor **18** can be arranged between the first resonator **14** and the second resonator **16** on the substrate **12**. The reference conductor **18** can be, for example, at the center between the first resonator **14** and the second resonator **16** on the substrate **12**, but the present disclosure is not limited thereto. For example, the reference conductor **18** may be at a position where the distance between the reference conductor **18** and the first resonator **14** differs from the distance between the reference conductor **18** and the second resonator **16**. The reference conductor **18** has an opening **18a**. The reference conductor **18** is configured to surround at least a part of the connection line path **20**.

(25) The connection line path **20** can be made of a conductor. The connection line path **20** is located between the first resonator **14** and the second resonator **16** in the Z-axis direction. The Z-axis direction can also be referred to as a first direction, for example. The connection line path **20** can be connected to each of the first resonator **14** and the second resonator **16**. The connection line path **20** can be configured integrally with the third resonator **22**. The connection line path **20** can be configured to be magnetically or capacitively connected to each of the first resonator **14** and the second resonator **16**, for example. For example, the connection line path **20** may be configured to be electrically connected to each of the first resonator **14** and the second resonator **16**. The connection line path **20** is connected to a side of the first resonator **14** parallel to the X-axis direction and is connected to a side of the second resonator **16** parallel to the X-axis direction. The connection line path **20** can be a path parallel to the Z-axis direction. The connection line path **20** can be a third resonator.

(26) In FIG. **1**, the connection line path **20** has been described as a linear path, but this is an example and does not limit the present disclosure. The connection line path **20** may include a plurality of paths such as a path parallel to the Z-axis direction and a path parallel to the XY plane.

(27) The third resonator **22** can be arranged between the first resonator **14** and the second resonator **16** in the Z-axis direction. The third resonator **22** can be inside the opening **18a** of the reference conductor **18**. The third resonator **22** can be inside the opening **18a** so as not to contact with the reference conductor **18**. The third resonator **22** can be configured to be magnetically or capacitively connected to each of the first resonator **14** and the second resonator **16**, for example. That is, the

third resonator **22** is surrounded by the reference conductor **18**. The third resonator **22** is capacitively connected to the reference conductor **18**.

(28) The feeder line **30** is electromagnetically connected to the first resonator **14**. The feeder line **30** is configured to supply power to the first resonator **14**. The input impedance of the feeder line **30** is, for example, 50Ω , but is not limited to this.

(29) In the present embodiment, when a wavelength of a fundamental wave of the arriving electromagnetic wave is λ , the length of at least one side of the first resonator **14** is set to $\lambda/2$, the length of at least one side of the second resonator **16** is set to $\lambda/2$, and the length of at least one side of the third resonator **22** is set to $\lambda/4$.

(30) In the present embodiment, the first resonator **14** is configured to transmit, to the feeder line **30**, the electromagnetic wave received from the Z-axis direction.

(31) The second resonator **16** is configured to resonate by a signal from the feeder line **30**. The second resonator **16** is configured to radiate an electromagnetic wave when resonated by the signal from the feeder line **30**. The second resonator **16** is configured to radiate the electromagnetic wave in the Z-axis direction when resonated by the signal from the feeder line **30**. The second resonator **16** is configured to transmit, to the feeder line **30**, the electromagnetic wave received from the -Z-axis direction.

(32) The first resonator **14** is configured to radiate an electromagnetic wave when resonated by a signal from the feeder line **30**. The first resonator **14** is configured to radiate the electromagnetic wave toward the Z-axis direction when resonated by the signal from the feeder line **30**.

(33) The second resonator **16** may be configured to resonate at a phase different from that of the first resonator **14** in response to the signal supplied from the feeder line **30**. The second resonator **16** may be configured to resonate in a direction different from the resonance direction of the first resonator **14** in the XY plane direction when resonated by the signal from the feeder line **30**. For example, when the first resonator **14** is configured to resonate in the X-axis direction, the second resonator **16** may be configured to resonate in the Y-axis direction.

(34) The first resonator **14** and/or the second resonator **16** may be configured such that the resonance direction changes over time in the XY plane direction.

(35) A radiation pattern of the antenna according to the first embodiment will be described with reference to FIG. 2. FIG. 2 is a view for illustrating the radiation pattern of the antenna according to the first embodiment.

(36) FIG. 2 illustrates a radiation pattern of an electromagnetic wave of the antenna **10** illustrated in FIG. 1. As illustrated in FIG. 2, the antenna **10** has large gains in the Z-axis direction and the -Z-axis direction. That is, the antenna **10** radiates the electromagnetic wave in the Z-axis direction and the -Z-axis direction.

(37) Frequency characteristics of the antenna according to the first embodiment will be described with reference to FIG. 3. FIG. 3 is a graph showing the frequency characteristics of the antenna according to the first embodiment.

(38) In FIG. 3, the horizontal axis represents the frequency [Giga Hertz (GHz)], and the vertical axis represents the gain [deci Bel (dB)]. FIG. 3 shows a graph G1. FIG. 3 shows a reflection coefficient of power supplied to the feeder line **30** of the antenna **10**. As shown in FIG. 3, the gain of the reflection coefficient is equal to or less than -5 dB from the vicinity of 18.00 GHz to the vicinity of 28.00 GHz. That is, in the antenna **10**, matching is achieved in a range from the vicinity of 18.00 GHz to the vicinity of 28.00 GHz.

(39) Radiation characteristics of the antenna according to the first embodiment will be described with reference to FIG. 4. FIG. 4 is a graph showing the radiation characteristics of the antenna according to the first embodiment.

(40) In FIG. 4, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dB]. FIG. 4 shows a graph G2 and a graph G3. The graph G2 shows radiation efficiency in the -Z-axis direction. The graph G3 shows radiation efficiency in the +Z-axis direction. As

shown in the graphs G2 and G3, the radiation efficiency is equal to or greater than -3 dB from the vicinity of 18.00 GHz to the vicinity of 28.00 GHz. The antenna **10** has good radiation characteristics in the $+Z$ -axis direction and the $-Z$ -axis direction.

(41) The peak gain of the antenna according to the first embodiment will be described with reference to FIG. 5. FIG. 5 is a graph showing the peak gain of the antenna according to the first embodiment.

(42) In FIG. 5, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dBi]. FIG. 5 shows a graph G4. As shown in FIG. 5, the peak gain is 4 dBi or lower from the vicinity of 18.00 GHz to the vicinity of 31.00 GHz. The antenna **10** has a good peak gain.

Second Embodiment

(43) A configuration example of an antenna according to a second embodiment will be described with reference to FIG. 6. FIG. 6 is a view illustrating a configuration example of the antenna according to the second embodiment.

(44) As illustrated in FIG. 6, an antenna **10A** includes a first resonator **14A**, a second resonator **16A**, the reference conductor **18**, a connection line path **20a**, a connection line path **20b**, a connection line path **20c**, a connection line path **20d**, the third resonator **22**, a first auxiliary reference conductor **24**, a second auxiliary reference conductor **26**, and the feeder line **30**.

(45) The first resonator **14A** is different from the first resonator **14** illustrated in FIG. 1 in that the length of at least one side is set to 214. The second resonator **16A** is different from the second resonator **16** illustrated in FIG. 1 in that the length of at least one side is set to 214.

(46) The first resonator **14A** is configured to resonate by receiving an electromagnetic wave from the $+Z$ -axis direction. The first resonator **14A** is configured to radiate the electromagnetic wave when resonating. The first resonator **14A** is configured to radiate the electromagnetic wave toward the $+Z$ -axis direction when resonating.

(47) The second resonator **16A** is configured to radiate the electromagnetic wave when resonating. The second resonator **16A** radiates the electromagnetic wave toward the $-Z$ -axis direction when resonating. The second resonator **16A** is configured to resonate by receiving the electromagnetic wave from the $-Z$ -axis direction.

(48) The second resonator **16A** may be configured to resonate at a phase different from that of the first resonator **14A**. The second resonator **16A** may be configured to resonate in a direction different from the resonance direction of the first resonator **14A** in the XY plane direction. For example, when the first resonator **14A** is configured to resonate in the X-axis direction, the second resonator **16A** may be configured to resonate in the Y-axis direction. The second resonator **16A** may be configured such that the resonance direction of the second resonator **16A** changes over time in the XY plane direction with respect to the resonance direction of the first resonator **14A**. The second resonator **16A** may be configured to attenuate a first frequency band of the electromagnetic wave received by the first resonator **14A** and radiate the resultant electromagnetic wave.

(49) The third resonator **22** can be arranged between the first resonator **14A** and the second resonator **16A** in the Z-axis direction. The third resonator **22** can be inside the opening **18c** of the reference conductor **18**. The third resonator **22** can be inside the opening **18c** so as not to contact with the reference conductor **18**. That is, the third resonator **22** is surrounded by the reference conductor **18**.

(50) The first auxiliary reference conductor **24** can be arranged between the first resonator **14A** and the reference conductor **18**. The first auxiliary reference conductor **24** can be made of a conductor. The second auxiliary reference conductor **26** can be arranged between the second resonator **16A** and the reference conductor **18**. The second auxiliary reference conductor **26** can be made of a conductor.

(51) One end of the connection line path **20a** is electromagnetically connected to the first resonator **14A**. The connection line path **20a** passes through the first auxiliary reference conductor **24**, and the other end of the connection line path **20a** is electrically connected to the reference conductor

18. The connection line path **20a** is electromagnetically connected to the first auxiliary reference conductor **24**. The connection line path **20a** can also be referred to as a first connection line path. (52) One end of each of the connection line path **20b**, the connection line path **20c**, and the connection line path **20d** is electromagnetically connected to the second resonator **16A**. The connection line path **20b**, the connection line path **20c**, and the connection line path **20d** pass through the second auxiliary reference conductor **26**, and the other end of each of the connection line path **20b**, the connection line path **20c**, and the connection line path **20d** is electromagnetically connected to the reference conductor **18**. The connection line path **20b**, the connection line path **20c**, and the connection line path **20d** are electromagnetically connected to the second auxiliary reference conductor **26**. Each of the connection line path **20b**, the connection line path **20c**, and the connection line path **20d** can also be referred to as a second connection line path.

(53) The feeder line **30** is electromagnetically connected to the first resonator **14A**. The feeder line **30** is configured to supply power to the first resonator **14**. The input impedance of the feeder line **30** is, for example, 50Ω , but is not limited to this.

(54) Frequency characteristics of the antenna according to the second embodiment will be described with reference to FIGS. 7 and 8. FIGS. 7 and 8 are graphs showing the frequency characteristics of the antenna according to the second embodiment.

(55) In FIG. 7, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dB]. FIG. 7 shows a graph G5. The graph G5 shows a reflection coefficient. For example, a gain of a frequency band in the vicinity of 19.00 GHz is about -9.4 dB. For example, a gain of a frequency band in the vicinity of 23.00 GHz is about -7.4 dB. For example, a gain of a frequency band in the vicinity of 26.00 GHz is about -19.9 dB.

(56) In FIG. 8, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dB]. FIG. 8 shows a graph G6 and a graph G7. The graph G6 shows radiation efficiency in the $-Z$ -axis direction. The graph G7 shows radiation efficiency in the $+Z$ -axis direction. As shown in the graphs G6 and G7, the radiation efficiency is equal to or greater than -3 dB from the vicinity of 19.00 GHz to the vicinity of 26.00 GHz. The antenna **10A** has good radiation characteristics in the $+Z$ -axis direction and the $-Z$ -axis direction.

(57) The peak gain of the antenna according to the second embodiment will be described with reference to FIG. 9. FIG. 9 is a graph showing the peak gain of the antenna according to the second embodiment.

(58) In FIG. 9, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dBi]. FIG. 9 shows a graph G8. As shown in FIG. 9, the peak gain is equal to or greater than -1 dBi from the vicinity of 19.00 GHz to the vicinity of 26.00 GHz. The antenna **10A** has good peak characteristics.

(59) A radiation pattern of the antenna according to the second embodiment will be described with reference to FIGS. 10, 11, and 12. FIGS. 10 to 12 are views for illustrating the radiation pattern according to the second embodiment.

(60) FIG. 10 illustrates a radiation pattern of the antenna **10A** at a frequency of 19 GHz. FIG. 11 illustrates a radiation pattern of the antenna **10A** at a frequency of 23 GHz. FIG. 12 illustrates a radiation pattern of the antenna **10A** at a frequency of 26 GHz. As illustrated in FIG. 10, when the frequency is 19 GHz, the maximum value of the gain is -0.5 dB, and the minimum value of the gain is -14.2 dB. As illustrated in FIG. 11, when the frequency is 23 GHz, the maximum value of the gain is 1.2 dB and the minimum value of the gain is -19.8 dB. As illustrated in FIG. 12, when the frequency is 26 GHz, the maximum value of the gain is 2.0 dB, and the minimum value of the gain is -27.5 dB.

OTHER EMBODIMENTS

(61) Frequency characteristics of an antenna according to other embodiments will be described with reference to FIG. 13. FIG. 13 is a graph showing the frequency characteristics of the antenna according to other embodiments.

(62) In FIG. 13, the horizontal axis represents the frequency [GHz], and the vertical axis represents the gain [dB]. FIG. 13 shows a graph G9. The graph G9 shows a reflection coefficient of a triple-band compatible antenna. For example, a gain of a frequency band in the vicinity of 19.00 GHz is about -1.5 dB. For example, a gain of a frequency band in the vicinity of 23.00 GHz is about -2 dB. For example, a gain of a frequency band in the vicinity of 26.00 GHz is about -3 dB.

(63) Embodiments of the present disclosure have been described above, but the present disclosure is not limited by the contents of the embodiments. Constituent elements described above include those that can be easily assumed by a person skilled in the art, those that are substantially identical to the constituent elements, and those within a so-called range of equivalency. The constituent elements described above can be combined as appropriate. Various omissions, substitutions, or modifications of the constituent elements can be made without departing from the spirit of the above-described embodiments.

Claims

1. An antenna comprising: a first resonator extending in a first plane direction; a second resonator spaced apart from the first resonator in a first direction and extending in the first plane direction; third resonator positioned between the first resonator and the second resonator in the first direction and configured to be magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator; a reference conductor extending in the first plane direction, positioned between the first resonator and the second resonator in the first direction, and serving as a potential reference of the first resonator and the second resonator; and a feeder line connected to the first resonator, wherein the reference conductor surrounds at least a part of the third resonator in the first plane direction.
2. The antenna according to claim 1, wherein the reference conductor comprises a through hole, and wherein the third resonator is magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator via the through hole.
3. The antenna according to claim 1, wherein the first resonator transmits, to the feeder line, an electromagnetic wave received from a forward direction of the first direction.
4. The antenna according to claim 1, wherein the second resonator radiates an electromagnetic wave when resonated by a signal from the feeder line.
5. The antenna according to claim 4, wherein the second resonator radiates an electromagnetic wave in a backward direction of the first direction when resonated by the signal from the feeder line.
6. The antenna according to claim 5, wherein the second resonator transmits, to the feeder line, an electromagnetic wave received from a backward direction of the first direction.
7. The antenna according to claim 1, wherein the first resonator radiates an electromagnetic wave when resonated by a signal from the feeder line.
8. The antenna according to claim 1, wherein the first resonator radiates an electromagnetic wave in a forward direction of the first direction when resonated by the signal from the feeder line.
9. The antenna according to claim 1, wherein the second resonator resonates at a phase different from a phase of the first resonator in response to the signal supplied from the feeder line.
10. The antenna according to claim 1, wherein the second resonator resonates, in the first plane direction, in an in-plane direction different from an in-plane direction of the first resonator when resonated by supply from the feeder line.
11. The antenna according to claim 1, wherein a resonance direction of the first resonator and/or the second resonator changes over time in the first plane direction.
12. An antenna comprising: a first resonator extending in a first plane direction; a second resonator spaced apart from the first resonator in a first direction and extending in the first plane direction; a reference conductor extending in the first plane direction, positioned between the first resonator

and the second resonator in the first direction, and serving as a potential reference of the first resonator and the second resonator; a third resonator positioned between the first resonator and the second resonator in the first direction and configured to be magnetically or capacitively connected to or electrically connected to each of the first resonator and the second resonator; a first auxiliary reference conductor positioned between the first resonator and the reference conductor and extending in the first plane direction; a second auxiliary reference conductor positioned between the second resonator and the reference conductor and extending in the first plane direction; a first connection line path configured to electromagnetically connect the first resonator, the reference conductor, and the first auxiliary reference conductor; and a second connection line path configured to electromagnetically connect the second resonator, the reference conductor, and the second auxiliary reference conductor, wherein the reference conductor surrounds at least a part of the third resonator in the first plane direction.

13. An array antenna comprising: one or more antennas according to claim 1, wherein the one or more antennas are arranged in the first plane direction.
